MA1521 L1 Week 10 Wed 20261021

Convert $\displaystyle\int_0^1\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{-(x^2+y^2)}^{(x^2+y^2)} 21xy^2\,dz\,dy\,dx$ to cylindrical coordinates and evaluate the new integral.

Cylindrical coordinates for triple integrals

$$dV_{(x,y,z)} = r\,dz\,dr\,d\theta$$

$\|$

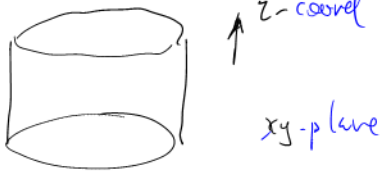
$dx\,dy\,dz$

$dz\,dy\,dx$

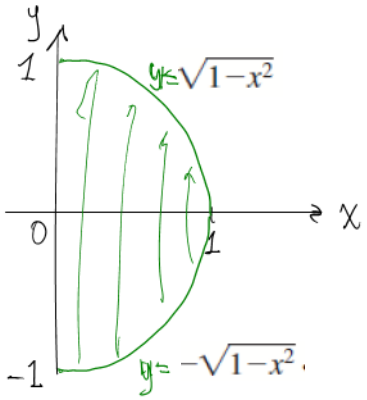
$dz\,dx\,dy$

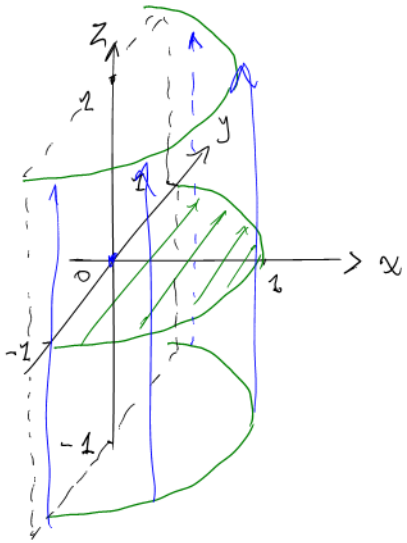
etc

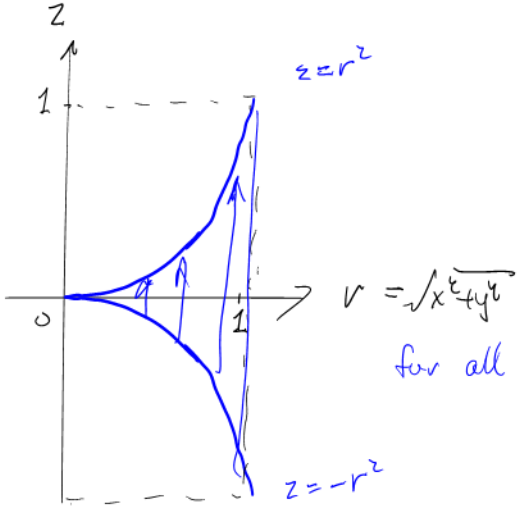
i.e. Cylindrical coordinates for x,y,z-space is "just" polar coordinates for x,y-plane together with z-coord



$$\int_{x=0}^{1}\int_{y=-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{z=-(x^2+y^2)}^{(x^2+y^2)} 21xy^2\,dz\,dy\,dx$$







for all angles $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$$\int_{x=0}^{1}\int_{y=-\sqrt{1-x^2}}^{y=\sqrt{1-x^2}}\int_{z=-(x^2+y^2)}^{z=(x^2+y^2)} 21xy^2\,dz\,dy\,dx$$

$$= \int_{\theta=-\frac{\pi}{2}}^{\theta=+\frac{\pi}{2}}\int_{r=0}^{r=1}\int_{z=-r^2}^{z=+r^2} \underbrace{21(r\cos\theta)(r\sin\theta)^2}_{21xy^2}\,r\,dz\,dr\,d\theta$$

cylindrical coords

$$= \int_{\theta=-\frac{\pi}{2}}^{\theta=+\frac{\pi}{2}}\int_{r=0}^{r=1} 21r^4\cos\theta\sin^2\theta\underbrace{\left[\int_{z=-r^2}^{z=+r^2} 1\,dz\right]}_{2r^2}dr\,d\theta$$

$$= 42\int_{\theta=-\frac{\pi}{2}}^{\theta=+\frac{\pi}{2}}\underbrace{\left[\int_{r=0}^{r=1} r^6\,dr\right]}_{\frac{r^7}{7}\Big|_0^1=\frac{1}{7}}\cos\theta\sin^2\theta\,d\theta$$

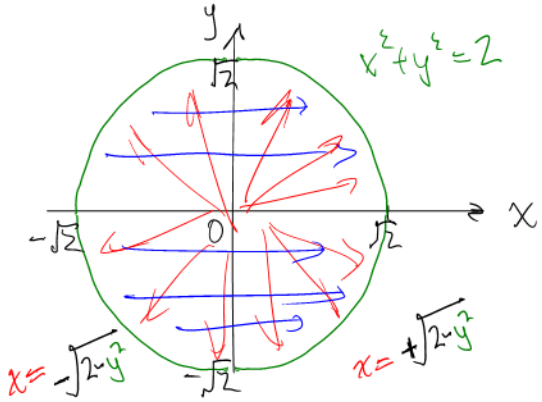
$$= 6\int_{\theta=-\frac{\pi}{2}}^{\theta=+\frac{\pi}{2}}\cos\theta\sin^2\theta\,d\theta = 6\left[\frac{\sin^3\theta}{3}\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 4 \quad \#$$

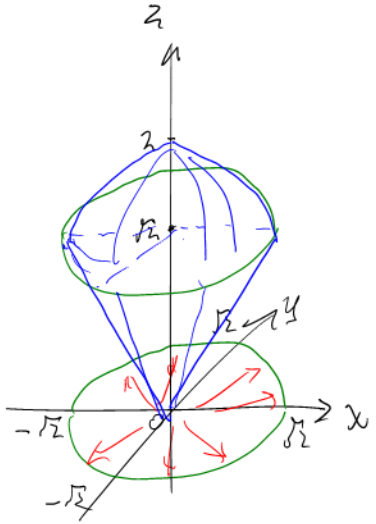
Cylindrical Coords

Convert $\int_0^{2\pi}\int_0^{\sqrt{2}}\int_r^{\sqrt{4-r^2}} 3\,dz\,r\,dr\,d\theta$ to rectangular coordinates with the order of integration $dz\,dx\,dy$ and evaluate the new integral.

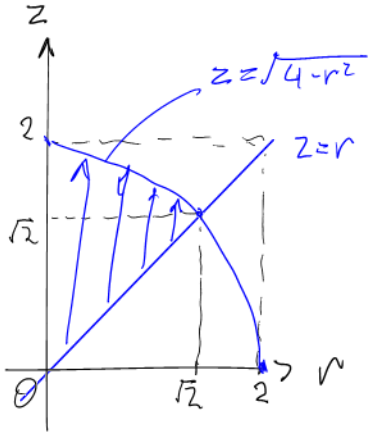
$\theta = 0 \to 2\pi$, $r = 0 \to \sqrt{2}$, $z = r \to \sqrt{4-r^2}$

$3\,dz\,r\,dr\,d\theta$





Fix $\theta \in [0, 2\pi]$



$\theta = 0 \to 2\pi$, $r = 0 \to \sqrt{2}$, $z = r \to \sqrt{4-r^2}$

$3\,dz\,r\,dr\,d\theta$ = cylind. coord differential

$= \int_{y=-\sqrt{2}}^{y=+\sqrt{2}} \int_{x=-\sqrt{2-y^2}}^{x=+\sqrt{2-y^2}} \int_{z=\sqrt{x^2+y^2}}^{z=\sqrt{4-x^2-y^2}} 3\,dz\,dx\,dy$

rect. coord differential

$$= \int_{y=-\sqrt{2}}^{y=+\sqrt{2}} \int_{x=-\sqrt{2-y^2}}^{x=+\sqrt{2-y^2}} 3\left(\sqrt{4-x^2-y^2} - \sqrt{x^2+y^2}\right) \, dx \, dy$$

carrying out $\int dx$ is super painful!

convert back to polar coords

$$= \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=\sqrt{2}} 3\left(\sqrt{4-r^2} - r\right) r \, dr \, d\theta$$

$$= \left(\int_{\theta=0}^{\theta=2\pi} 1 \, d\theta\right) \cdot \left(\int_{r=0}^{r=\sqrt{2}} 3\left(\sqrt{4-r^2} - r\right) r \, dr\right)$$

2π

$$\left[-(4-r^2)^{3/2} - r^3\right]_{r=0}^{r=\sqrt{2}}$$

$$= -2\sqrt{2} + 8 - 2\sqrt{2} + 0$$

$$= 8 - 4\sqrt{2}$$

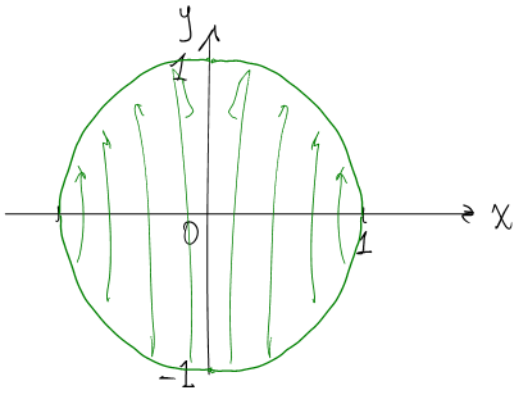
$$= 2\pi\left(8 - 4\sqrt{2}\right) \quad \#$$

Evaluate: $\iiint_B z\,dx\,dy\,dz$ where B is the region within the cylinder $x^2 + y^2 = 1$

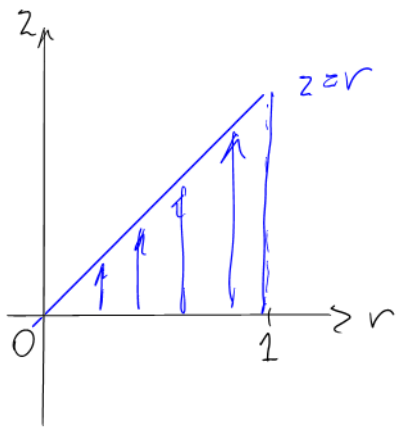
$z \geq 0$ above the xy plane

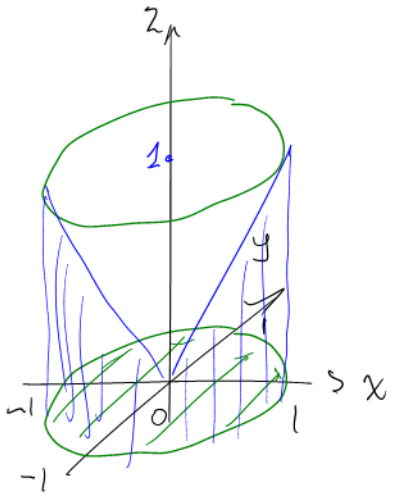
and below the cone $z = (x^2 + y^2)^{1/2}$.

$z = r$



Fix any $\theta \in [0, 2\pi]$





solid cylinder with (inverted) cone hollowed out

$$\iiint_B z\,dx\,dy\,dz = \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=1} \int_{z=0}^{z=r} z \; r\,dz\,dr\,d\theta$$

↑ use cylindrical coords

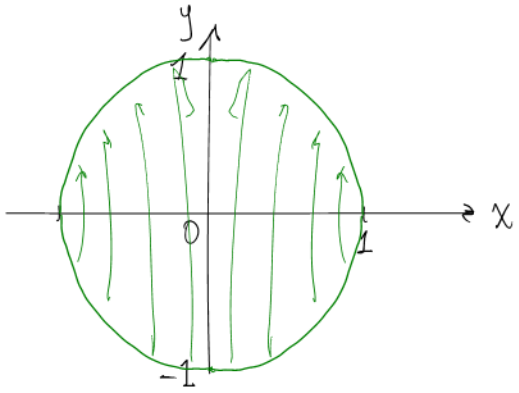
$$= \left[\int_{r=0}^{r=1} r \left(\int_{z=0}^{z=r} z\,dz\right) dr\right] \left(\int_{\theta=0}^{\theta=2\pi} d\theta\right)$$

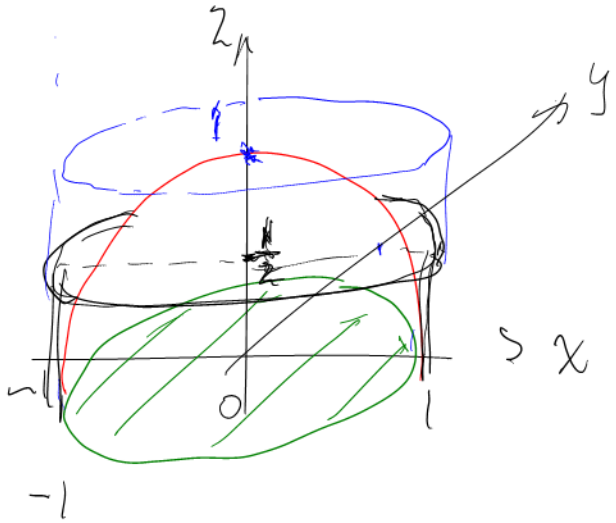
$$= \left(\int_{r=0}^{r=1} \frac{r^3}{2}\, dr \right) \cdot 2\pi = \left[\frac{r^4}{8} \right]_0^1 \cdot 2\pi = \frac{\pi}{4}$$

#

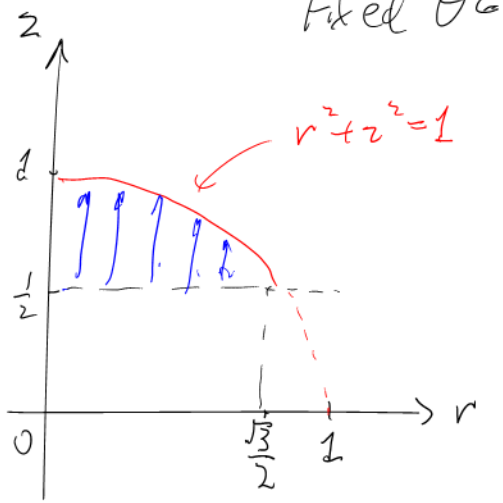
Evaluate: $\iiint_W (x^2+y^2+z^2)^{-1/2}\,dx\,dy\,dz$ where W is the region determined by $\frac{1}{2} \le z \le 1$ and $x^2+y^2+z^2 \le 1$.

lies within unit sphere





Fixed $\theta \in [0, 2\pi]$



$$\iiint_W (x^2+y^2+z^2)^{-1/2}\,dx\,dy\,dz = \int_{\theta=0}^{\theta=2\pi} \int_{r=0}^{r=\frac{\sqrt{3}}{2}} \int_{z=\frac{1}{2}}^{z=\sqrt{1-r^2}} (r^2+z^2)^{-1/2}\, r\, dz\, dr\, d\theta$$

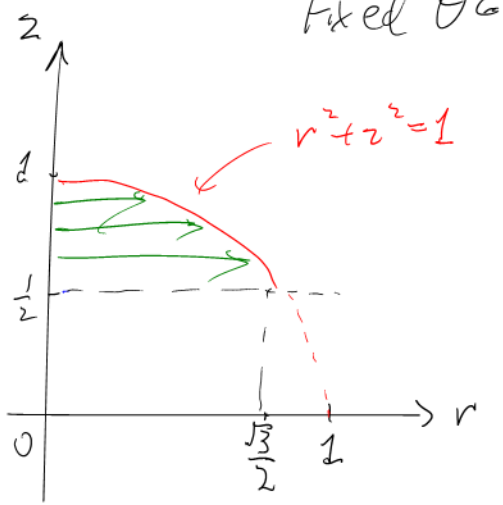
use cylindrical coords

Need to do this integration (try!)

(to be continued ... !)

Alternative: still use cylindrical coords, but with the order $r\,dr\,dz\,d\theta$

$$\iiint_W (x^2+y^2+z^2)^{-1/2}\,dx\,dy\,dz =$$



Fixed $\theta \in [0, 2\pi]$

$$= \int_{\theta=0}^{\theta=2\pi}\int_{z=\frac{1}{2}}^{z=1}\int_{r=0}^{r=\sqrt{1-z^2}} \left(r^2+z^2\right)^{-1/2} r\,dr\,dz\,d\theta$$

$$= \int_{\theta=0}^{\theta=2\pi}\int_{z=\frac{1}{2}}^{z=1}\int_{r=0}^{r=\sqrt{1-z^2}} \frac{r}{\sqrt{r^2+z^2}}\,dr\,dz\,d\theta$$

$$= \int_{\theta=0}^{\theta=2\pi}\int_{z=\frac{1}{2}}^{z=1} \left[\sqrt{r^2+z^2}\right]_{r=0}^{r=\sqrt{1-z^2}} dz\,d\theta = \int_{\theta=0}^{\theta=2\pi}\int_{z=\frac{1}{2}}^{z=1} (1-z)\,dz\,d\theta$$

$$= \int_{\theta=0}^{\theta=2\pi} \left[z-\frac{z^2}{2}\right]_{z=\frac{1}{2}}^{z=1} d\theta$$

$$= \int_{\theta=0}^{\theta=2\pi} \left(\frac{1}{2}-\frac{1}{2}\left(\frac{3}{4}\right)\right) d\theta = 2\pi\cdot\frac{1}{8} = \frac{\pi}{4}$$